

Advanced (Habanero)

Q1. Simplify $\sqrt{16} + \sqrt{9}$

- (A) 5
- (B) 7
- (C) 3
- (D) 9
- (E) 1

Q2. Simplify $\sqrt{25} \cdot \sqrt{4}$

- (A) 10
- (B) 20
- (C) 5
- (D) 15
- (E) 30

Q3. Simplify $\sqrt{9} - \sqrt{4}$

- (A) 1
- (B) 3
- (C) 5
- (D) 2
- (E) 0

Q4. Simplify $\frac{\sqrt{16}}{\sqrt{4}}$

- (A) 2
- (B) 4
- (C) 8
- (D) 1
- (E) 6

Q5. Simplify $\sqrt{36} + \sqrt{9}$

- (A) 6
- (B) 9
- (C) 3
- (D) 12
- (E) 15

Q6. Simplify $\sqrt{49} \cdot \sqrt{9}$

- (A) 21
- (B) 63
- (C) 42
- (D) 35
- (E) 45

Q7. Simplify $\sqrt{64} - \sqrt{16}$

- (A) 4
- (B) 6
- (C) 2
- (D) 8
- (E) 0

Q8. Simplify $\frac{\sqrt{25}}{\sqrt{25}}$

- (A) 0
- (B) 1
- (C) 2
- (D) 5
- (E) 10

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{12} + \sqrt{27}$ and show your work

Q10. Simplify $\frac{\sqrt{48}}{\sqrt{12}}$ and show your work

Chef's Tips

- Tip 1: Check for any common factors in the radicand before simplifying
- Tip 2: Rationalize the denominator, if necessary, after simplifying
- Tip 3: Use the properties of radicals to simplify expressions